Why did one alpha in eight thousand come back?

? A pudding atom should never turn a fast alpha around. What did the bounces actually prove?

1

Predict first, then look

Thomson spread the positive charge through a sphere of radius about 10^{-10} m. An alpha inside it feels a field growing linearly from the centre and peaking at kZe/R² at the surface. Work out the impulse before looking at the data: that is what makes this a test.

$$\theta \sim \frac{kZe^2}{KR}, \qquad \theta \lesssim 1^\circ$$

→ multiple scattering averages out further: the pudding predicts a straight beam

2

The impulse budget of a soft charge

A 5 MeV alpha spends about 10^{-18} s inside the atom, and the largest available force in that time gives a transverse momentum far below its own: a fraction of a degree. To turn it 90° the charge must sit inside 10^{-14} m, where the force is a hundred times larger.

$$f = \pi b^2 nt: \quad b_{90^\circ} \approx 2.3 \times 10^{-14}\ \text{m} \ \Rightarrow\ f \approx \frac{1}{8000}$$

→ f is the fraction of nuclei's cross-sectional area the foil presents

3

What the one-in-eight-thousand means

The observed fraction is an area measurement. Invert f = πb²nt and the angle-90° impact parameter falls out: nucleus tens of fm, atom 10^5 fm — a marble in an empty cathedral.

→ r_min is an upper bound on nuclear size while the force stays Coulomb — it is not a radius

4

A worked closest approach

A 5 MeV alpha on gold, head-on. All kinetic energy becomes potential energy at the turning point:

$$r_{\min} = \frac{2(79)(1.44\ \text{eV nm})}{5 \times 10^6\ \text{eV}} = 4.55 \times 10^{-5}\ \text{nm}$$

$$r_{\min} = 45.5\ \text{fm} \quad \text{against } R \approx 7\ \text{fm}$$

→ six nuclear radii of clearance: the alpha never touched the nucleus, so the Coulomb law held

KEY RESULT

the two numbers

$$r_{\min} = \frac{2kZe^2}{K}$$

$$\tan\frac{\theta}{2} = \frac{kZe^2}{mv^2 b}$$

f = πb²nt is an area ratio, not a force

TRAP

what r_min is not

It is not the nuclear radius. It is a head-on upper bound, valid only while the force is pure Coulomb.

Large-angle scattering does not mean contact: it means small b.

NUMBERS

carry these

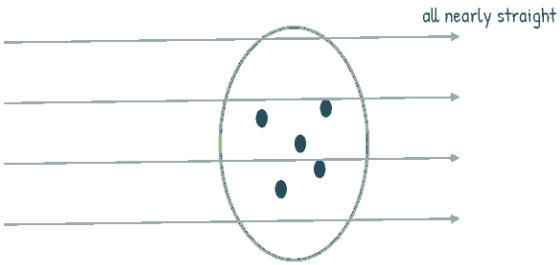
1 in 8000 back-scattered (gold, ~90°)

kZe² for gold = 114 eV nm

nuclear scale few fm; atom 10^5 fm

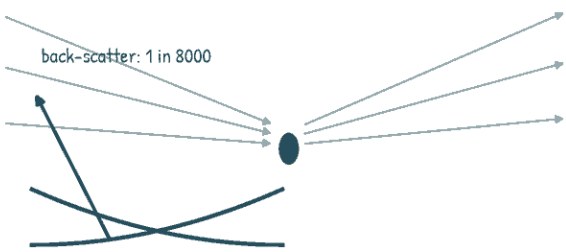
Same beam, two predictions: the rare back-scatter is the test that decides

a soft cloud cannot turn a fast alpha



Thomson: charge spread over 10^-10 m

a point charge can turn it around



Rutherford: one point charge, sin^-4 law

KEEP / the rare back-scatter is the measurement: charge sits in a nucleus, not a cloud

How close can an alpha get before the law breaks?

? Rutherford's formula held at every angle measured. Where must it stop, and what stops it?

1 The hyperbola, in one line

A Coulomb encounter is a conic section. The impact parameter b is the miss distance the alpha would have had with no force at all; the scattering angle follows from angular momentum plus the impulse integral.

$$\tan\frac{\theta}{2}=\frac{kZe^2}{mv^2 b}=\frac{r_h}{2b},\qquad r_h=\frac{2kZe^2}{K}$$

→ b → 0 gives back-scatter; b → ∞ gives a straight track

2 Invert it and you have a ruler

Because the angle depends on b only through the ratio r_h/2b, measuring θ measures b. The fraction scattered beyond a given angle is then the area of the disk πb² times the areal density of nuclei, nt.

$$\frac{d\sigma}{d\Omega}=\left(\frac{r_h}{4}\right)^2\frac{1}{\sin^4(\theta/2)}$$

→ the $\sin^{-4}$ law: halving the angle multiplies the count sixteenfold

3 Where the Coulomb picture must fail

Set r_min equal to the nuclear radius. Above that beam energy the alpha touches the nuclear surface, the potential stops being 1/r, and the measured distribution departs from $\sin^{-4}$. Natural alphas at 5–8 MeV probe only the clean Coulomb regime; accelerators above a few tens of MeV see the deviation.

$$K_{\text{fail}}=\frac{2kZe^2}{R}\approx 32\ \text{MeV}\quad(\text{gold},\ R\approx 7\ \text{fm})$$

4 A worked 30 MeV probe

Raise the beam to 30 MeV on the same gold foil and see how much room is left:

$$r_{\min}=\frac{227.5\ \text{eV nm}}{3\times10^7\ \text{eV}}=7.6\times10^{-6}\ \text{nm}$$

$$=7.6\ \text{fm}\approx R_{\text{Au}}$$

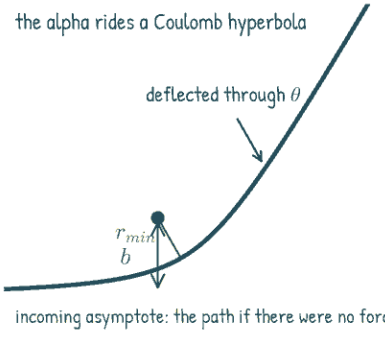
→ at about 32 MeV the alpha touches the surface and $\sin^{-4}$ scattering starts to fail

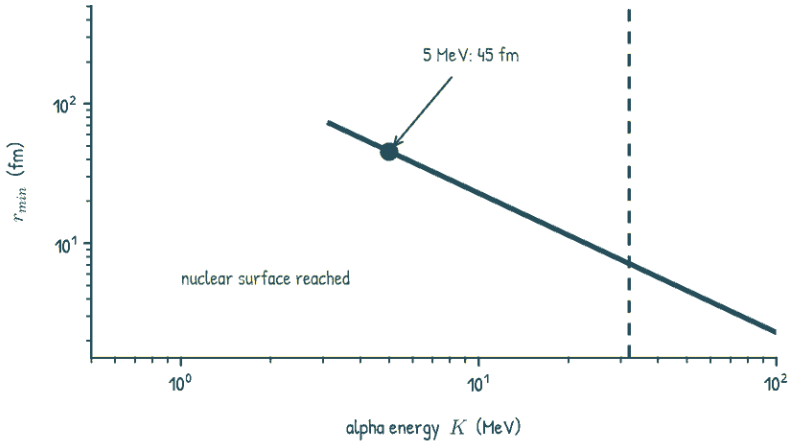
KEY RESULT
the geometry
$\tan(\theta/2)=r_h/2b$
$r_h=2kZe^2/K$
At 90°: b = r_h/2, always.

GATE
when the formula holds
Pure Coulomb, point nucleus, no recoil, non-relativistic alpha.
All four fail at high energy or high Z.

TRAP
angle convention
θ is measured from the incoming direction, not from the normal to the foil.

r_min is a head-on upper bound on the nuclear size, not a radius measurement





KEEP / r_min shrinks as 1/K until it reaches the nuclear surface — that is where new physics starts

▼13.6 eV
0.1 eV 10 MeV

Why is angular momentum n times h-bar?

? Bohr postulated it. de Broglie's wave makes it a closure condition on a circle.

1

A wave on a ring either closes or cancels

Put a de Broglie wave on a circle. If a whole number of wavelengths fits the circumference it meets itself in phase and stands; if not, it destroys itself. That single condition is the quantisation.

$$2\pi r = n\lambda = n\frac{h}{p} \;\Rightarrow\; pr = n\hbar$$

→ on a circle, pr is exactly the angular momentum — nothing else had to be assumed

2

Add the force, get the atom

Two equations now: closure and Coulomb force balance. Eliminate v between them and every hydrogen number follows with no adjustable constant.

$$\frac{mv^2}{r} = \frac{ke^2}{r^2}, \qquad mvr = n\hbar$$

$$r_n = \frac{n^2\hbar^2}{m_e k e^2} = 0.529\,\frac{n^2}{Z}\ \text{Å}$$

3

The same algebra gives speed and energy

Substituting back gives the speed and, with E = K + U, the ladder. K = −E and U = 2E, so the virial relation is a free check on every result.

$$v_n = \frac{kZe^2}{n\hbar} = 2.19\times10^6\,\frac{Z}{n}\ \text{m/s}$$

$$E_n = -\frac{m_e k^2 Z^2 e^4}{2\hbar^2 n^2} = -13.6\,\frac{Z^2}{n^2}\ \text{eV}$$

→ K = −E and U = 2E, so the electron is always more kinetic than bound

4

A worked wavelength check

The n = 3 orbit of hydrogen, checked against de Broglie directly:

$$v_3 = 7.29\times10^5\ \text{m/s}, \qquad \lambda = \frac{h}{m_e v_3} = 9.97\ \text{Å}$$

$$3\lambda = 29.9\ \text{Å} = 2\pi(4.76\ \text{Å})\ \checkmark$$

→ a 10 % error in n would leave the wave 36 degrees out of phase, cancelling itself

KEY RESULT

the closure

$$2\pi r = n\lambda$$

$$L = n\hbar$$

One condition replaces a postulate.

INSIGHT

why it is exact

The condition fixes the shape of the wave, and the Coulomb force fixes its scale. Two equations, two unknowns, no fitting.

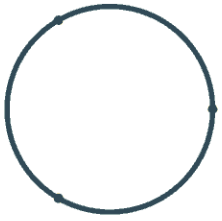
TRAP

the two-pi slip

L = nh/(2π) = nℏ, never nh. And λ = h/p, not h/mv with a different v.

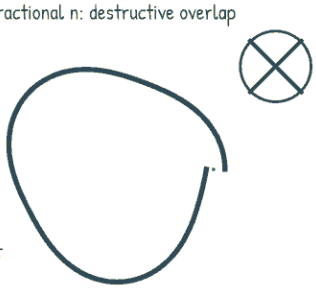
Bohr quantisation is a standing-wave condition: L = n h-bar is 2 pi r = n lambda

3 whole wavelengths close in phase



$2\pi r = n\lambda$

fractional n: destructive overlap



no closed wave, no orbit

KEEP / an orbit is allowed when the electron's wave closes on itself in phase

▼13.6 eV
0.1 eV 10 MeV

What does the hydrogen ladder actually look like?

? Thirty years of spectroscopy in one diagram. Read it as a table, not a picture.

① Three columns, one formula each

Every hydrogenic quantity is a power of n. The radius grows as n², the speed falls as 1/n, and the binding energy falls as 1/n². Write them as three columns and the page becomes a lookup table rather than a derivation.

$$r_n = 0.529\frac{n^2}{Z}\ \text{Å}, \quad v_n = 2.19\times10^6\frac{Z}{n}, \quad E_n = -13.6\frac{Z^2}{n^2}\ \text{eV}$$

② The gap, not the level, is what you see

A photon carries the difference between two levels. Using hc = 1240 eV nm, a gap in eV converts to a wavelength by one division, which is the whole arithmetic of the chapter.

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right), \qquad \lambda = \frac{1240}{\Delta E}$$

→ R = 1.097×10⁷ m⁻¹; hcR = 13.6 eV, so the two forms are the same statement

③ Bound means negative

Zero energy is the electron at rest at infinity. A bound electron therefore has negative total energy, and its ionisation energy is the positive quantity −E_n. Excitation energy is the difference of two levels; ionisation always ends at zero.

$$\text{IE}_n = -E_n = \frac{13.6\,Z^2}{n^2}\ \text{eV}, \qquad \text{IP (volts)} = \text{IE (eV)}$$

→ for a singly charged electron the two numbers coincide; for an alpha they do not

④ A worked ionisation ladder

Ionise hydrogen from n = 2, and then from n = 3:

$$\text{IE}_2 = \frac{13.6}{4} = 3.40\ \text{eV}, \qquad \text{IE}_3 = \frac{13.6}{9} = 1.51\ \text{eV}$$

$$\lambda_{\text{edge}} = \frac{1240}{3.40} = 365\ \text{nm}$$

→ the Balmer edge at 365 nm is the same number seen as a wavelength

KEY RESULT
the ladder

$$E_n = -13.6\,Z^2/n^2$$

$$\lambda = 1240/\Delta E\ (\text{nm, eV})$$

r ∝ n², v ∝ 1/n, |E| ∝ 1/n²

CHECK
instant audits

Ground state must return −13.6 eV at Z = 1.

A gap must be smaller than the ionisation energy it sits under.

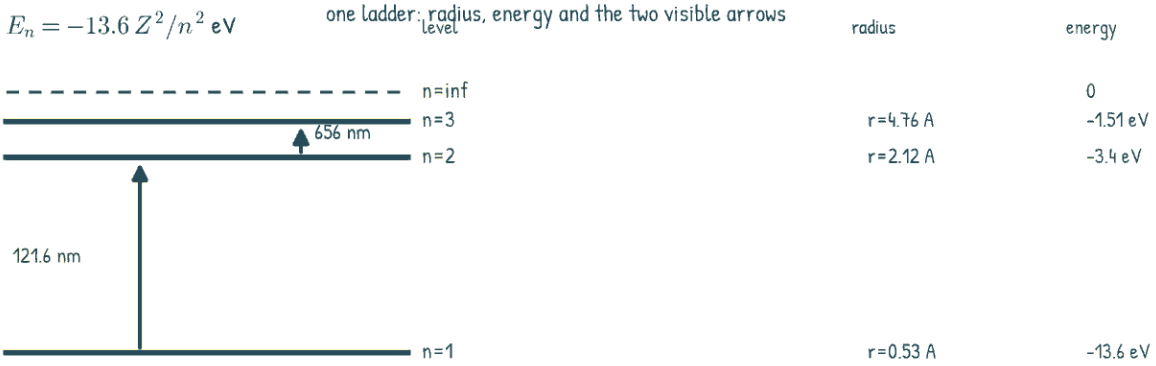
TRAP
signs and units

Subtracting two excited levels when asked to ionise — ionisation ends at 0 eV.

Plugging a wavelength in nm into hc = 1240 eV nm the wrong way round.

One formula per column: radius grows as n squared, binding falls as 1/n squared

$E_n = -13.6\,Z^2/n^2$ eV



KEEP / radius as n², binding as 1/n², and 13.6(1/n_f² − 1/n_i²) eV per arrow

▼1.89 eV
0.1 eV 10 MeV

Which series is which, and why five names?

? Lyman, Balmer, Paschen, Brackett, Pfund: one formula, five finishing rungs.

① **Fix the lower level and the family appears**

Everything that lands on n = 1 is Lyman (ultraviolet), on n = 2 is Balmer (the visible one), on n = 3 is Paschen (infrared), then Brackett and Pfund further out. The formula does not change; only n_f does.

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), \qquad n_i > n_f$$

② **The two ends of any series**

The longest wavelength in a series is the smallest step, n_f + 1 → n_f. The shortest is the limit n_i → ∞, which is the ionisation edge of that lower level. Every series therefore runs from a line to a continuum edge.

$$\lambda_{\max} = \frac{1}{R}\frac{n_f^2(n_f+1)^2}{2n_f+1}, \qquad \lambda_{\min} = \frac{n_f^2}{R}$$

→ Balmer: 656 nm down to the 365 nm edge, which is why it is the visible one

③ **What the eye can and cannot see**

Only Balmer overlaps the 380–750 nm window, and only its first three lines are comfortably visible. Lyman is vacuum ultraviolet; Paschen is infrared. This is not an accident of hydrogen — it is why stellar spectra are classified by which series is strong.

→ a hydrogen discharge looks pink because Hα at 656 nm dominates the visible output

④ **A worked series limit**

The two ends of the Balmer series, computed from the same rule:

$$\lambda_{\max} = \frac{1}{R}\frac{4\cdot 9}{5} = 656 \text{ nm } (n_i = 3)$$

$$\lambda_{\min} = \frac{4}{R} = 364.6 \text{ nm } (n_i \to \infty)$$

→ everything between is visible; beyond 364.6 nm the series ends in a continuum

KEY RESULT
the five names
Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5.
Only Balmer is visible.

$$\lambda_{\min} = n_f^2/R$$

TRAP
counting the wrong thing
"Lines in the Balmer series from n = 5" is four (5→2, 4→2, 3→2 plus the 5→… no: 5→2, 4→2, 3→2 = three). Total lines from n = 5 is ten.

NUMBERS
bench numbers
Hα 656.3 nm · Hβ 486.1 · Hγ 434.0 · Hδ 410.2
Lyman α 121.6 nm · Balmer edge 364.6 nm

Every series is the same Rydberg rule with a different finishing rung



one ladder, three series, one visible window



KEEP / the series is named by its lower level: n_f = 1, 2, 3, 4, 5

What can a 12.5 eV electron do that a 12.5 eV photon cannot?

? Same energy, different menu. The rule about who may pay in part is worth marks every year.

1 ## The electron menu

A bombarding electron can transfer any amount up to its whole kinetic energy, because it keeps the remainder as its own motion. So every level with excitation energy below the beam energy is reachable.

$$E_{\text{exc}} = 13.6\left(1 - \tfrac{1}{n^2}\right)\text{eV}: \;\; 10.2,\;\; 12.09,\;\; 12.75$$

→ a 12.5 eV beam reaches n = 2 and n = 3, and stops short of n = 4

2 ## The photon menu

A photon is created or absorbed as one unit. It cannot give 10.2 eV and carry 2.3 eV away, so a 12.5 eV photon is not absorbed by the 10.2 eV transition at all — it simply passes. Resonance is a matching condition, not a threshold.

$$hf = E_i - E_f \quad \text{exactly}$$

→ this is why the absorption spectrum has dark lines at precise wavelengths only

3 ## Above the ionisation limit

Once the photon energy exceeds the binding energy the electron is freed and the surplus becomes kinetic energy. The spectrum changes character: discrete lines below the limit, a continuum above it. The ionisation edge is the last line of a series.

$$K_{\max} = hf - \text{IE}, \qquad hf > \text{IE}$$

4 ## A worked threshold ladder

Which levels can a 12.5 eV electron reach in hydrogen?

$$E_2 = 10.2, \quad E_3 = 12.09, \quad E_4 = 12.75 \text{ eV}$$

$$12.5 > 12.09 \text{ but } 12.5 < 12.75 \;\Rightarrow\; n \le 3$$

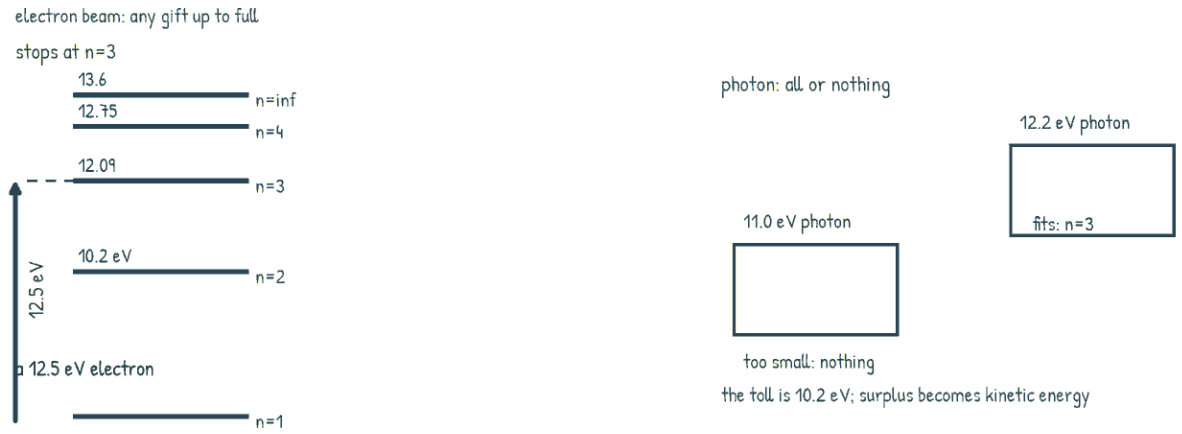
→ and the surplus 0.41 eV stays with the electron as kinetic energy

KEY RESULT
the asymmetry
$$E_{\text{exc}} = 13.6(1 - n^{-2})$$
Electron: any gift up to full. Photon: all or nothing.

TRAP
three classics
Letting an 11 eV photon excite the 10.2 eV level.
Forgetting the cascade: exciting n = 3 gives three lines, not one.
Confusing excitation energy with ionisation energy.

NUMBERS
hydrogen thresholds
n=2: 10.2 eV / 121.6 nm
n=3: 12.09 eV / 102.6 nm
ionisation: 13.6 eV / 91.2 nm

Electrons may pay in part; photons must pay in full - that is the excitation asymmetry



KEEP / electrons may hand over part of their energy; photons must be absorbed whole

How far does Z² scaling carry you?

? He^+, Li^{2+}, muonic hydrogen and positronium are the same atom with two numbers changed.

1

Charge: divide the radius, square the energy

A nucleus of charge Ze pulls harder, so the orbit shrinks as 1/Z while every energy multiplies by Z². He^+ therefore has a ground state at −54.4 eV and an orbit of 0.265 Å; Li^{2+} has −122.4 eV.

$$r_n \propto \frac{1}{Z}, \qquad E_n \propto Z^2, \qquad \lambda \propto \frac{1}{Z^2}$$

→ the 2 → 1 line of Li^{2+} is 13.5 nm, deep in the soft X-ray

2

Mass: the nucleus is not nailed down

The electron and nucleus orbit their common centre of mass, and every mass-dependent result takes the reduced mass μ = m_eM/(m_e+M). For hydrogen the correction is 0.05 %; for deuterium it shifts H^α by 0.18 nm, which is how deuterium was found.

$$R_M = R_\infty \frac{\mu}{m_e}, \qquad \Delta\lambda/\lambda \approx \frac{m_e}{2M}$$

3

Exotic atoms are scaling tests

Replace the electron with a muon and μ ≈ 207 m_e: the atom shrinks by 207 and binds 207 times harder, so its orbit sits inside the nucleus and its levels measure nuclear size. Positronium has μ = m_e/2, so it is twice as big and half as bound.

$$a_\mu = \frac{0.529\ \text{Å}}{207} = 256\ \text{fm}, \qquad E_1 = -2.81\ \text{keV}$$

→ positronium: 1.06 Å, −6.8 eV, and every wavelength doubled

4

A worked Li2+ line

The 2 → 1 resonance line of Li^{2+} (Z = 3):

$$\Delta E = 13.6(9)\left(\tfrac{1}{1} - \tfrac{1}{4}\right) = 91.8\ \text{eV}$$

$$\lambda = \frac{1240}{91.8} = 13.5\ \text{nm}$$

→ soft X-ray, and exactly one ninth of hydrogen's 121.6 nm: λ ∝ 1/Z²

KEY RESULT

the two dials

$$r \propto n^2/Z$$

$$E \propto Z^2/n^2$$

and m_e → μ for any mass-dependent quantity

TRAP

forgetting Z

Using −13.6/n² for He^+. Every energy carries Z², every radius 1/Z.

Using m_e where the reduced mass belongs — the whole point of exotic atoms.

NUMBERS

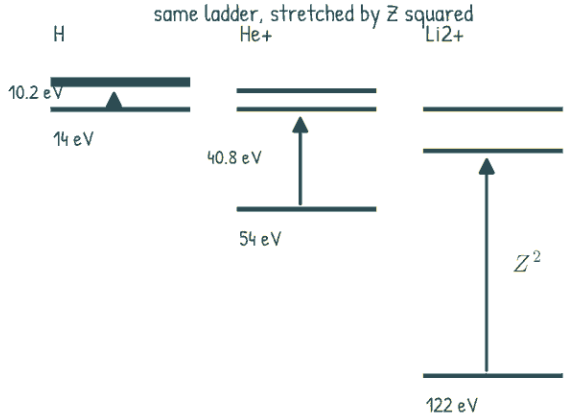
scaled ground states

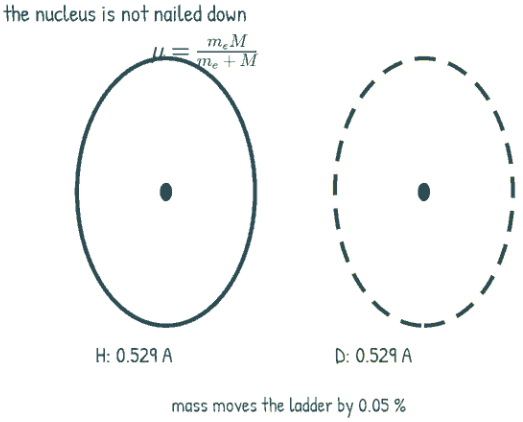
H −13.6 eV, 0.529 Å

He^+ −54.4 eV, 0.265 Å

Li^{2+} −122.4 eV, 0.176 Å

Z squared deepens the ladder; the reduced mass shifts every rung by a part in 10^4





KEEP / charge scales everything by Z², mass by the reduced mass — nothing else changes

▼1.89 eV
0.1 eV 10 MeV

What are the other three numbers for?

? Bohr gave one integer. Spectroscopy needs four, and each one explains a different splitting.

1 The numbers and their counts

For a given n, l runs 0…n−1, m_l runs −l…+l, and m_s is ±½. Counting states gives 2n² per shell, which is the periodic-table arithmetic in one line.

$$n:\ 1, 2,\ \ldots;\quad l < n;\quad -l \le m_l \le l;\quad m_s = \pm\tfrac{1}{2}$$

→ shell n holds 2n² states: 2, 8, 18, 32

2 A field turns one level into a fan

An orbiting electron is a current loop with magnetic moment μ_B per unit ħ. A field B splits a level into 2l+1 components spaced by μ_B B, which is why a line seen in a magnetic field becomes a triplet.

$$\Delta E = m_l \mu_B B, \qquad \mu_B = 5.79 \times 10^{-5}\ \mathrm{eV/T}$$

→ at 1 T the Hα line splits by 0.02 nm: small, but resolvable with a grating

3 Selection rules decide which lines exist

Not every pair of levels radiates: a photon carries one unit of angular momentum, so Δl = ±1 and Δm_l = 0, ±1. That is why some transitions are bright, others forbidden, and why the number of lines from a cascade is smaller than the number of level pairs.

$$\Delta l = \pm 1, \qquad \Delta m_l = 0,\ \pm 1$$

4 A worked cascade count

An atom in n = 5 cascades to the ground state. How many spectral lines, and how many from n = 5 alone?

$$N = \frac{5 \cdot 4}{2} = 10 \text{ lines}$$

$$\text{from } n = 5 \text{ only}:\ 5 \to 4, 3, 2, 1 = 4$$

→ the series count is n − n_f; the cascade count is all pairs

KEY RESULT
counts and splits

$$N = n(n-1)/2$$

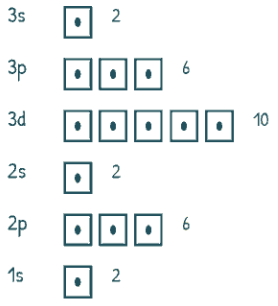
$$\Delta E = m_l \mu_B B$$

2n² states in shell n

GATE
what this model ignores
Spin-orbit fine structure (10^{-4} eV) is named, not derived.
The anomalous factor g ≈ 2 is outside the syllabus.

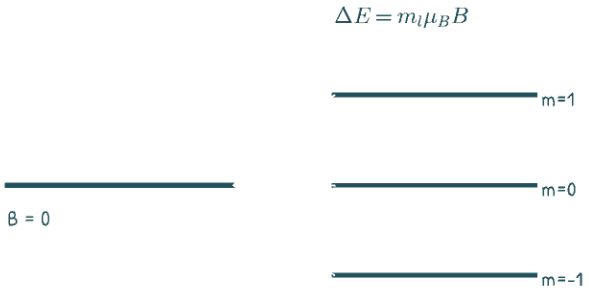
TRAP
reading the question
If it says "orbital only", use m_lμ_BB and do not double for spin.

The Bohr ladder is one quantum number; the other three decide which lines exist



orbitals e⁻ slots

n, l, m_l, m_s: 2n^2 states in shell n



a magnetic field fans one level into three

KEEP / n sets the energy; l, m_l, m_s decide which transitions are allowed and how they split

What exactly is in a beam of light?

? Counting coins: energy per photon, photons per second, and the push they give.

1

Energy per photon

One quantum carries hf, which in the units of this chapter is 1240 divided by the wavelength in nanometres. That single conversion does most of the numerical work in the chapter.

$$E = hf = \frac{hc}{\lambda} = \frac{1240}{\lambda\ [\text{nm}]}\ \text{eV}$$

→ green light at 550 nm: 2.25 eV; a 100 keV X-ray: 0.0124 nm

2

Photons per second, and per cubic metre

Divide the beam power by the energy per photon. Spread over a sphere, the intensity falls as 1/r², so the flux does too; inside the beam the number density is the flux divided by c.

$$\dot{N} = \frac{P}{hf}, \qquad n = \frac{I}{hf}, \qquad n_{\text{vol}} = \frac{I}{hfc}$$

→ a 100 W sodium lamp at 589 nm emits 3×10²⁰ photons per second

3

A beam pushes

Each photon arrives with momentum h/λ and leaves with either none (absorbed) or the reverse (reflected). The force is the momentum delivered per second, which is P/c or 2P/c. This is radiation pressure, derived here by counting rather than by fields.

$$F = \frac{P}{c}(1+R), \qquad p_{\text{rad}} = \frac{I}{c}\,(1+R)$$

→ the force from a 1 W reflected beam is 6.7 nN — a microgram's weight

4

A worked photon count

A 5 mW laser at 633 nm: how many photons leave per second, and what force does the beam exert if absorbed?

$$E = \frac{1240}{633} = 1.96\ \text{eV} = 3.14 \times 10^{-19}\ \text{J}$$

$$\dot{N} = \frac{5 \times 10^{-3}}{3.14 \times 10^{-19}} = 1.6 \times 10^{16}\ \text{s}^{-1}$$

→ F = P/c = 1.7×10⁻¹¹ N — a piconewton, and optical tweezers use exactly this

KEY RESULT

the three lines

$$E = 1240/\lambda\ (\text{eV, nm})$$

$$p = h/\lambda$$

$$F = P(1+R)/c$$

TRAP

massive-photon thinking

Using p = E/c for an electron. That relation is for massless particles; electrons take p = √(2mK).

Forgetting the factor 2 when a surface reflects.

NUMBERS

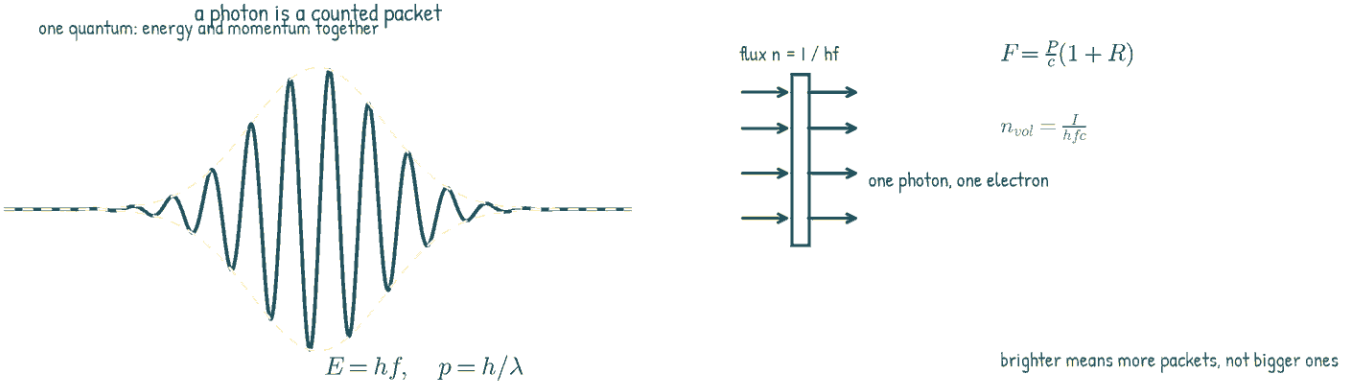
one-line conversions

hc = 1240 eV nm = 1.99×10⁻²⁵ J m

1 eV = 1.602×10⁻¹⁹ J

h = 6.63×10⁻³⁴ J s

Photon bookkeeping: energy per packet, and packets per second



KEEP / a beam is n photons per second, each carrying hf and momentum h/λ

▼1.8 eV
0.1 eV 10 MeV

Why does brighter light not give faster electrons?

? Five experimental facts, and the two that no wave theory can survive.

1 The five facts

There is a threshold frequency below which nothing is emitted at any intensity. Emission is instantaneous, within about a nanosecond. Current is proportional to intensity. Maximum kinetic energy grows with frequency and ignores intensity. The stopping potential is therefore a function of frequency alone.

→ three of the five are also what a wave predicts — they are decoys, not discriminators

2 Einstein's one line

One photon is absorbed by one electron. The electron pays a toll φ to escape the surface and keeps the rest as kinetic energy. Nothing about intensity appears here, which is why the maximum energy ignores it.

$$K_{\max} = hf - \phi, \qquad eV_s = K_{\max}$$

→ if hf < φ the right-hand side is negative and the answer is "no emission"

3 The graph that measured Planck's constant

Plot stopping potential against frequency. Every metal gives the same slope h/e and a different intercept −φ/e. Millikan's measurements of that slope agreed with the blackbody value of h, which is the cross-check that made the photon unavoidable.

$$V_s = \frac{h}{e}f - \frac{\phi}{e}, \qquad f_0 = \frac{\phi}{h}$$

→ slope 4.14×10^{-15} V s for every metal ever measured

4 A worked zinc threshold

Zinc has φ = 4.3 eV. What is the longest wavelength that ejects electrons, and what happens at 250 nm?

$$\lambda_0 = \frac{1240}{4.3} = 288 \text{ nm}$$

$$250 \text{ nm}: \; K_{\max} = \frac{1240}{250} - 4.3 = 4.96 - 4.3 = 0.66 \text{ eV}$$

→ visible light (400–700 nm) never discharges zinc: the photons are simply too poor

KEY RESULT
the balance

$$K_{\max} = hf - \phi$$

$$eV_s = hf - \phi$$

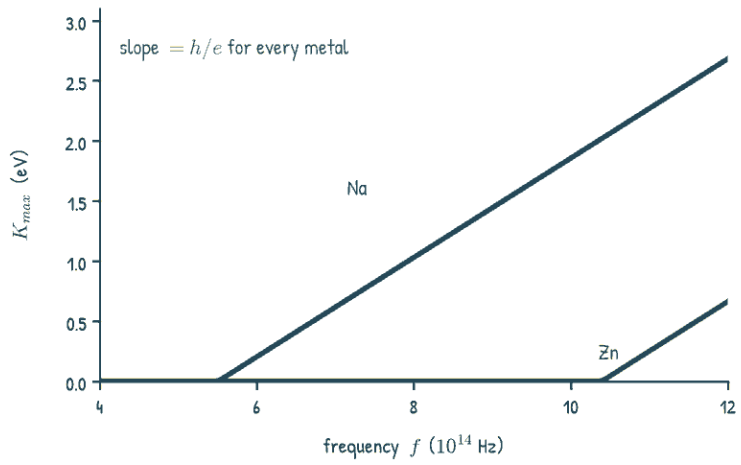
threshold $\lambda_0 = 1240/\varphi$ (nm, eV)

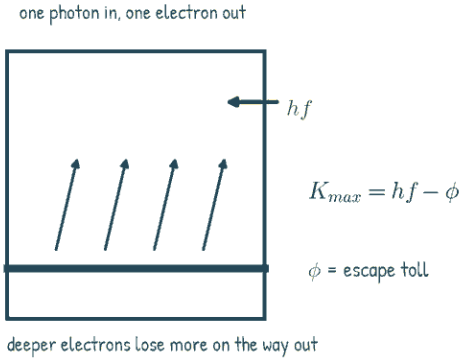
TRAP
four automatic zeros
Brighter light raising V_s.
Writing K_max = hf and forgetting the toll.
Expecting emission at threshold with energy: at λ_0 electrons emerge with zero.

NUMBERS
work functions
Na 2.28 eV · Zn 4.3 eV · Cs 1.9 eV · Pt 5.6 eV

Einstein equation: the photon pays a toll, the rest becomes kinetic energy

K_max = hf - phi





KEEP / intensity sets the number of electrons; frequency sets their maximum energy

How do you read a photocurrent graph?

? Two plateaux, one cutoff. Learn which knob moves which feature.

1 One curve, three features

Sweep the collector voltage. At large negative voltage the current is zero — even the fastest electron is turned back. Raise the voltage, electrons are collected, and at large positive voltage every emitted electron is collected: the current saturates.

$$eV_s = K_{\max}, \qquad I_{\text{sat}} \propto \text{ emission rate}$$

→ the saturation plateau is a collection limit, not a quantum effect

2 Doubling the intensity

Twice the photons means twice the electrons, so the plateau doubles. But each electron still gets the same photon energy, so the cutoff stays exactly where it was. Two curves, one edge.

→ any answer that moves the cutoff with intensity contradicts the whole experiment

3 Raising the frequency instead

At fixed intensity, higher frequency means fewer photons per second — each carries more energy — so the current actually falls slightly while the cutoff climbs. This is the subtle row that examiners use to separate people who track photons from people who track watts.

$$n = \tfrac{I}{hf}: \quad f\uparrow \;\Rightarrow\; n\downarrow \text{ at fixed } I$$

4 A worked graph reading

A V_s against f line passes through (5.0×10¹⁴ Hz, 0) and (11.0×10¹⁴ Hz, 2.48 V).

$$\text{slope} = \frac{2.48}{6.0\times10^{14}} = 4.13\times10^{-15}\ \text{V s}$$

$$h = e(4.13\times10^{-15}) = 6.62\times10^{-34}\ \text{J s}, \quad \phi = hf_0 = 2.07\ \text{eV}$$

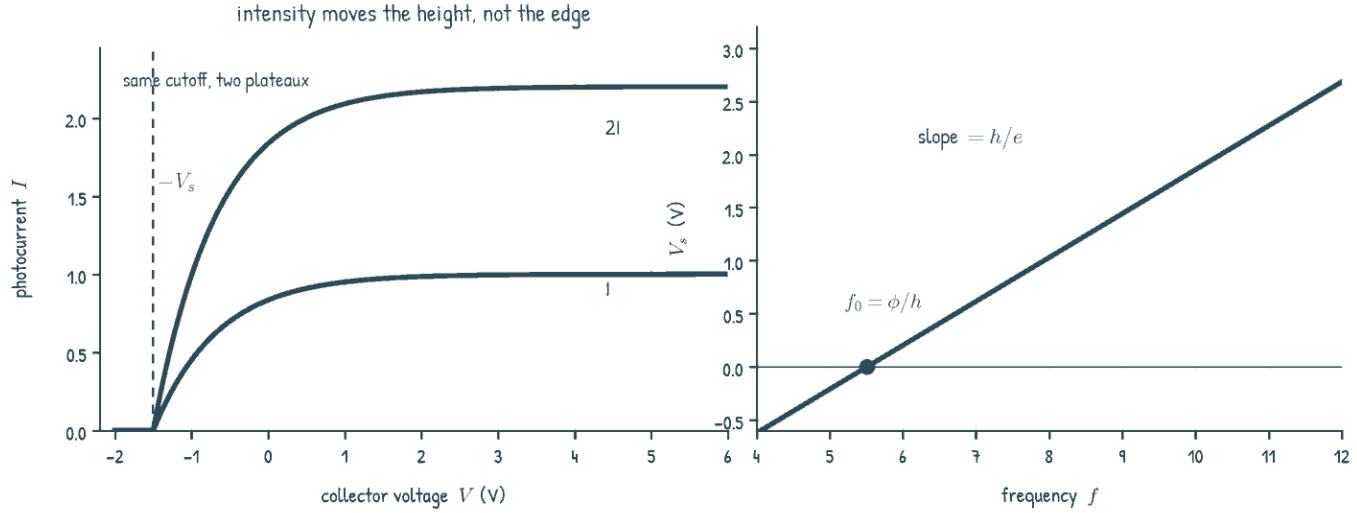
→ the slope must be h/e = 4.14×10⁻¹⁵ V s for every metal ever measured

KEY RESULT
read the graph
$eV_s = K_{\max}$
Left intercept: −V_s. Plateau height: rate. Slope of V_s vs f: h/e.

GATE
assumptions
One photon per electron; clean surface with a single φ; no field emission; no space charge.

TRAP
saturation is not quantum
Citing the saturation current as evidence for photons. It is geometry — every emitted electron collected.

Plot V_s against f: slope h/e, intercept −phi/e – Millikan measured h this way



KEEP / the cutoff marks the fastest electron; the plateau marks how many there are

What does a photocell measure, and what does it not?

? A battery and an ammeter separate two quantities that classical physics could not.

1 The apparatus is a filter

An evacuated tube holds an emitter and a collector. Light ejects electrons from the emitter; a variable voltage either helps them across or turns them back. The ammeter reads how many arrive, the voltmeter tells you at what retarding voltage they stop.

→ evacuated matters: in air the electrons collide with molecules within micrometres

2 Three knobs, three different answers

Intensity, frequency and collector voltage each move one feature and leave the others alone. That independence is the experimental content of the photon model, and it is why the photocell is a measuring instrument rather than a demonstration.

$$I_{\text{sat}} \propto I, \qquad V_s = \frac{hf - \phi}{e}$$

3 Where the ideal picture leaks

Real photocells have a spread of work functions, so the knee is soft rather than sharp. Very high intensity brings space-charge limiting; very high frequency makes the window absorb. None of these changes the threshold logic; they change the shape of the knee.

→ a soft knee is not evidence against quantisation — it is a dirty surface

4 A worked quantum efficiency

A photocell at 400 nm receives 1 mW and delivers 20 µA.

$$\dot{N}_\gamma = \frac{10^{-3}}{4.97 \times 10^{-19}} = 2.0 \times 10^{15}\ \text{s}^{-1}$$

$$\dot{N}_e = 1.25 \times 10^{14}\ \text{s}^{-1} \ \Rightarrow\ \eta = 6.2\ \%$$

→ a few per cent is normal for a plain metal; caesiated surfaces reach tens of per cent

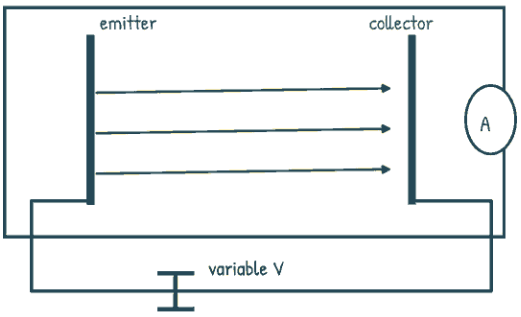
KEY RESULT
what each knob does
intensity → plateau
frequency → cutoff
collector voltage → collection only

INSIGHT
why it matters
The same device is a light meter, a night-vision tube and a solar cell's front end: one photon in, one electron out.

TRAP
current versus frequency
At fixed intensity, higher f gives fewer photons and a slightly lower current — not a higher one.

The photocell separates rate (intensity) from energy (frequency) with one battery

one battery measures both cutoff and rate

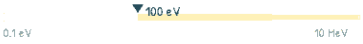


knob	current	V_s
intensity up	up	same
frequency up	down a little	up
voltage up	saturates	same

fixed I, higher f: fewer, richer photons

so the current dips while the cutoff climbs

KEEP / the battery measures energy per electron; the ammeter measures electrons per second



Do electrons have a wavelength?

? If light can be a particle, matter can be a wave. The only question is how big.

1 The hypothesis, and its one-line consequence

Attach to any particle the same relation that photons obey. For an electron accelerated from rest through V volts, the kinetic energy fixes the momentum, and the wavelength follows.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e eV}} = \frac{1.226}{\sqrt{V}}\ \text{nm}$$

→ 100 V gives 0.123 nm — the spacing of atoms, which is why crystals diffract electrons

2 Why you never see a cricket ball diffract

A 0.15 kg ball at 30 m/s has $\lambda \approx 10^{-34}$ m. Wave behaviour appears only when the wavelength is comparable to the geometry it meets, so a ball needs slits 10^{-34} m apart. The classical world is the limit $\lambda \ll$ every dimension.

$$\lambda_{\text{ball}} = \frac{6.63 \times 10^{-34}}{(0.15)(30)} = 1.5 \times 10^{-34}\ \text{m}$$

3 Electron microscopes are the practical payoff

Resolution is limited by the wavelength used. At 10 kV an electron has λ = 0.012 nm, four orders finer than green light, so the same lens geometry can in principle resolve viral structure. Aberrations eat most of that gain; they do not remove it.

→ slow electrons (tens of eV) have wavelengths of a few ångström — ideal for surfaces

4 A worked wavelength pair

Electrons at 100 V and at 10 kV:

$$\lambda_{100} = \frac{1.226}{\sqrt{100}} = 0.123\ \text{nm}$$

$$\lambda_{10^4} = 0.0123\ \text{nm}$$

→ a hundredfold voltage gives exactly a tenfold shorter wave: $\lambda \propto V^{-1/2}$

KEY RESULT

the bridge

$$\lambda = h/p$$

$$\lambda = 1.226/\sqrt{V}\ \text{nm}$$

p = √(2mK) non-relativistic; p = √(K² + 2mKc²)/c exact

TRAP

the four percent

The square-root formula is wrong above about 3 kV by more than 1 %.

And p = E/c is only for massless particles.

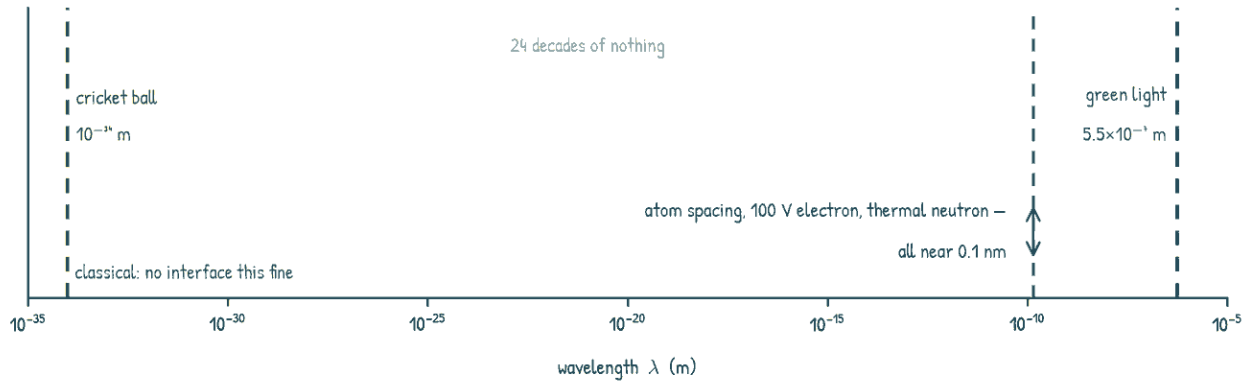
NUMBERS

bench wavelengths

100 V: 0.123 nm · 10 kV: 0.0123 nm

thermal neutron: 0.18 nm · green light: 550 nm

The same formula covers a cricket ball and an electron; only the ratio to the slit decides



KEEP / λ = h/p: large for electrons at a few volts, absurdly small for anything you can hold

Was the electron wave ever actually seen?

? Two experiments in 1927 turned a hypothesis into a measurement: a crystal and a detector.

1

A crystal is a grating with ångström spacing

Atoms in a crystal sit 1–3 Å apart, which is the wavelength of a 50–150 V electron. The crystal therefore acts as a three-dimensional grating, and reflected beams appear only at angles where the path difference between successive planes is a whole number of wavelengths.

$$2d\sin\theta = n\lambda$$

→ θ is the glancing angle, measured from the plane — not from the normal

2

The 1927 numbers

Davisson and Germer fired 54 V electrons at nickel and found a strong lobe at 65° from the incident beam. With the known plane spacing of 0.91 Å, Bragg predicts a wavelength of 1.65 Å; de Broglie gives 1.67 Å. The two agree to the accuracy of the day.

$$\lambda_{\text{Bragg}} = 2(0.91)\sin 65^\circ = 1.65\ \text{Å}$$

$$\lambda_{\text{deB}} = \frac{1.226}{\sqrt{54}} = 1.67\ \text{Å}$$

3

What followed

The same logic runs today as LEED in every surface lab, and as electron diffraction in structural biology. Slow electrons are used deliberately: a long wavelength both diffracts strongly and penetrates only a few atomic layers, so the pattern is a picture of the surface.

→ slower electrons diffract more, not less — lower energy means longer wavelength

4

A worked angle

54 V electrons on nickel planes of 0.91 Å:

$$\lambda = 1.67\ \text{Å}, \quad \sin\theta = \frac{1.67}{1.82} = 0.918$$

$$\theta = 66^\circ \quad \text{observed } 65^\circ$$

→ the one-degree gap is the rounding of d in the textbook value, not a failure of the theory

KEY RESULT

the pair

$$2d\sin\theta = n\lambda$$

$$\lambda = h/p$$

Bragg gives the wavelength from geometry; de Broglie predicts it from voltage.

GATE

why the peak is sharp

Constructive interference over thousands of planes; a small angle error destroys the sum.

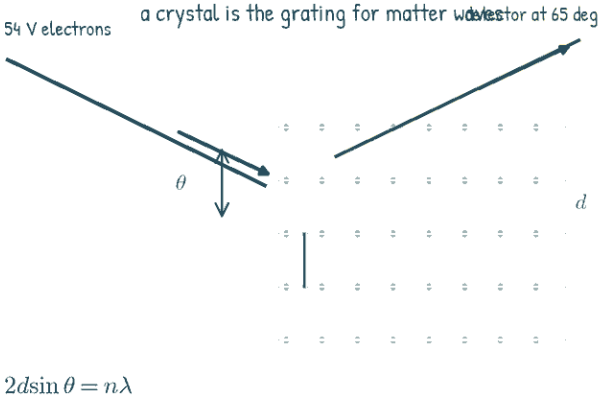
TRAP

angle and order

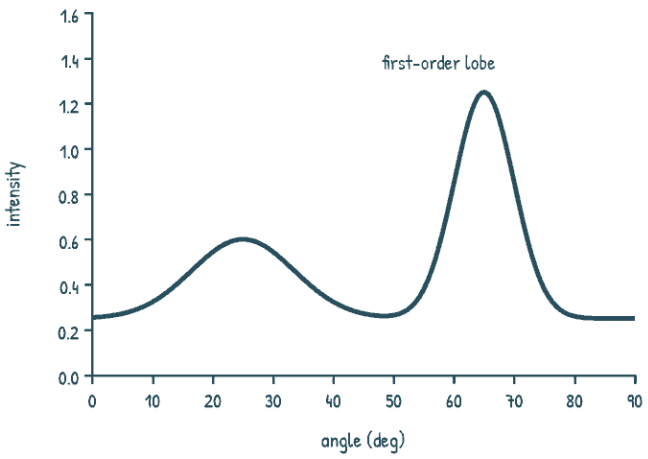
Writing 2d cosθ, or calling the second observed peak n = 1.

Forgetting that the detector angle is 2θ.

Davisson-Germer: the 65 degree lobe sits where 2d sin(theta) = lambda predicts



$2d\sin\theta = n\lambda$



KEEP / the diffraction angle follows 2d sinθ = nλ, and λ is exactly h/p

▼13.6 eV
0.1 eV 10 MeV

Why does the electron not fall into the nucleus?

? A classical orbiting charge radiates and dies in picoseconds. Something holds the atom up.

1 A localised particle cannot have a definite momentum

A pure sine has a perfect wavelength and no position. To localise a particle you superpose many waves; the narrower the packet, the wider the spread of momenta.

$$\Delta x\,\Delta p \gtrsim \frac{\hbar}{2}$$

→ a statement about waves, not about clumsy equipment; squeeze further and the kinetic term wins

2 Minimise the total energy

Confine an electron to radius r: its momentum is at least $\hbar/r$, so the kinetic term grows as $1/r^2$ while the Coulomb well deepens as $-1/r$. The sum has a minimum, and that minimum is the hydrogen atom.

$$E(r) \approx \frac{\hbar^2}{2m_e r^2} - \frac{ke^2}{r}$$

$$r_{\min} = \frac{\hbar^2}{m_e k e^2} = 0.529\ \text{Å}, \qquad E_{\min} = -13.6\ \text{eV}$$

3 Two more payoffs you will use

A particle in a small box keeps a minimum energy even at zero temperature — why helium stays liquid and nucleons stay energetic. And a state living only τ seconds has an energy width $\hbar/\tau$, the natural width of every spectral line.

$$K_{\min} \approx \frac{\hbar^2}{2mL^2}, \qquad \Delta E \sim \frac{\hbar}{\tau}$$

4 A worked ground state

Evaluate the two terms of E(r) at r = 0.5 Å:

$$K = \frac{\hbar^2}{2m_e r^2} = 15.3\ \text{eV}, \qquad U = -\frac{ke^2}{r} = -28.8\ \text{eV}$$

$$E = 15.3 - 28.8 = -13.5\ \text{eV}$$

→ the exact answer is −13.6 eV: the estimate misses by one per cent, and the physics is all there

KEY RESULT
the three uses
$\Delta x \Delta p \gtrsim \hbar/2$

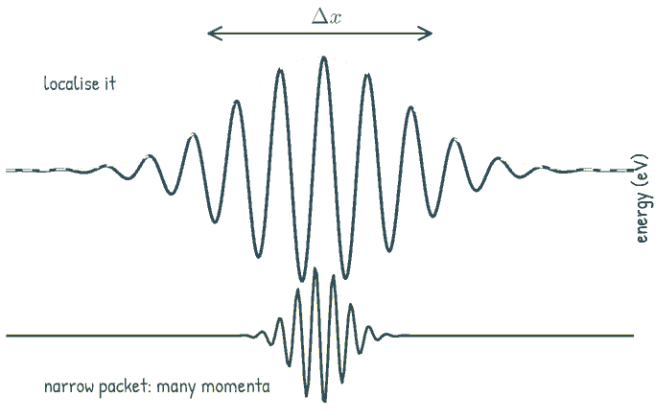
$E(r) = \hbar^2/2mr^2 - ke^2/r$

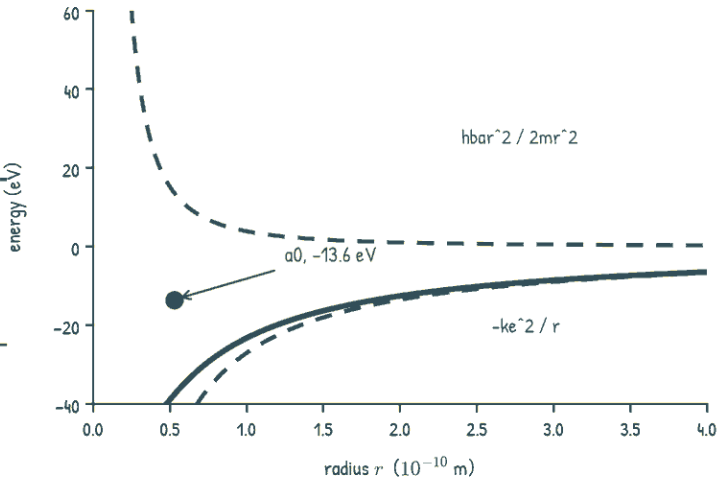
Confinement, zero-point energy, natural linewidth.

INSIGHT
why it is exact here
For the hydrogen ground state the estimate saturates the bound, so it returns a_0 and −13.6 eV exactly. Do not expect that for every state.

TRAP
the missing half
Writing $\Delta x \Delta p \geq \hbar$, or L = nh. The bound is $\hbar/2$ and L = $n\hbar$.

Squeezing costs kinetic energy: the sum has a floor, and that floor is the atom





KEEP / confinement costs momentum; the ground state is where that cost stops paying

0.1 eV 1 MeV 10 MeV

Why does every nucleus have the same density?

? Add nucleons and the ball grows, but the stuff inside never gets denser. What fixes the size?

1 A saturated force makes a fixed-volume ball

The strong force is attractive at 1–2 fm, gone by about 3 fm, and charge-independent. Because it is short-range, each nucleon binds only its neighbours: doubling A doubles the binding, not the binding per nucleon. Constant energy per nucleon means constant volume per nucleon.

$$V = Av = \tfrac{4}{3}\pi R^3 \;\Rightarrow\; R = R_0 A^{1/3}, \qquad R_0 = 1.2 \text{ fm}$$

→ the cube root is the whole content: density, not radius, is the invariant

2 Read the density off the constant

Substituting the fitted radius gives a density with no A in it. That is why the answer to "carbon or uranium?" is "neither, they are the same" — a fact worth carrying into any estimate.

$$\rho = \frac{3m_N}{4\pi R_0^3} = 2.3 \times 10^{17} \text{ kg/m}^3$$

→ a 5 cm³ teaspoon of it weighs 1.2×10¹² kg — a mountain in a spoon

3 What the numbers actually are

Carbon-12 has R = 2.75 fm, uranium-238 has R = 7.44 fm: only 2.7 times bigger for twenty times the nucleons. The nucleus occupies 10^{-12} of the atom's volume, yet holds 99.97 % of its mass.

$$R_{12} = 1.2(12)^{1/3} = 2.75 \text{ fm}, \qquad R_{238} = 1.2(238)^{1/3} = 7.44 \text{ fm}$$

→ the electron cloud is 10^5 times wider, and almost all of the atom is empty

4 A worked uranium radius

Uranium-238: radius and the mass of a 5 cm³ spoonful.

$$R = 1.2(238)^{1/3} = 7.44 \text{ fm}, \qquad V = \tfrac{4}{3}\pi R^3 = 1.72 \times 10^{-42} \text{ m}^3$$

$$m = \rho V = (2.3 \times 10^{17})(5 \times 10^{-6}) = 1.2 \times 10^{12} \text{ kg}$$

→ a billion tonnes in a teaspoon — the number that makes the density believable

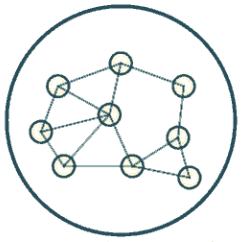
KEY RESULT
size and density
$R = 1.2A^{1/3}$ fm
$\rho = 2.3 \times 10^{17}$ kg/m³
Same for every nucleus.

GATE
where the law is soft
Liquid-drop mean radius. Halo nuclei and the surface fuzz bend it at the few-percent level.
It says nothing about shape: deformed nuclei have two radii.

TRAP
the two confusions
Saying density falls as A grows (it does not), and taking R ∝ A instead of A^1/3.

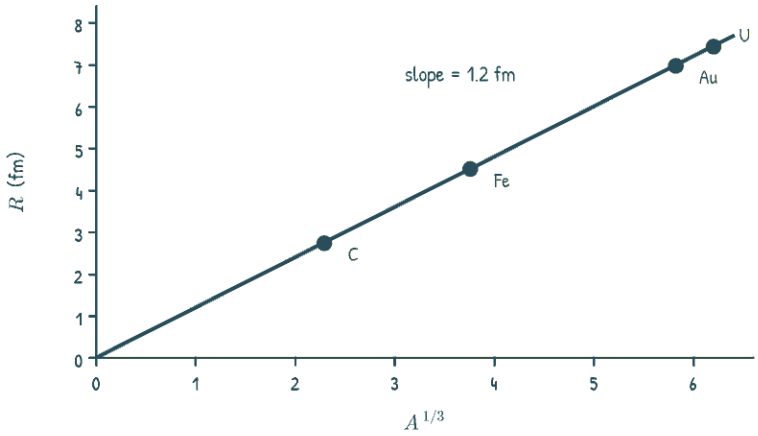
Saturated binding gives every nucleon the same volume: R = 1.2 A^(1/3) fm, density fixed

saturated attraction binds neighbours only



Coulomb repulsion: every proton pair, no saturation

radius grows as the cube root of A



KEEP / R = 1.2 A^1/3 fm is the statement that nuclear matter is incompressible

0.1 eV 100 MeV 10 MeV

Where does the missing mass go?

? Weigh a nucleus and its parts: the whole is lighter. The difference is what holds it together.

1

Weigh the parts, weigh the whole

Take Z protons and N neutrons apart, then assemble them. The assembly is lighter by Δm, and that mass went out as the energy that bound it. Conversely, Δm c² is the energy you must pay to take it apart again.

$$B = [Zm_H + Nm_n - M_{\text{atom}}]c^2, \qquad 1\ \text{u} = 931.5\ \text{MeV}$$

→ atomic masses are used throughout: the electron bookkeeping cancels for the nucleus

2

Two numbers set the scale

The deuteron is barely bound: 2.22 MeV for two nucleons, 1.11 each. Helium-4 is the first tight package: 28.3 MeV, 7.07 per nucleon — a factor 6.4 jump that explains why fusing past hydrogen pays so much.

$$\Delta m_d = 0.002388\ \text{u} \Rightarrow B = 2.22\ \text{MeV}, \qquad B/A = 1.11$$

$$\Delta m_{He} = 0.030377\ \text{u} \Rightarrow B = 28.3\ \text{MeV}, \qquad B/A = 7.07$$

3

The curve is the decision-maker

Plot B/A against A and every energy question becomes "which way does this move the nucleons?". Fusion climbs the left slope toward iron; fission climbs the right slope toward the same summit. Anything that moves nucleons up the curve releases the difference.

$$\text{fission}: \ 7.6 \to 8.5\ \text{MeV/nucleon} \ \Rightarrow \ \sim 200\ \text{MeV}$$

→ iron has nothing to gain — hence the endpoint of stellar burning

4

A worked mass defect

Helium-4 from its parts: 2 hydrogen atoms and 2 neutrons.

$$\Delta m = 2(1.007825) + 2(1.008665) - 4.002603 = 0.030377\ \text{u}$$

$$B = 0.030377(931.5) = 28.30\ \text{MeV}, \qquad B/A = 7.07\ \text{MeV}$$

→ the deuteron gives 1.11 per nucleon: that factor 6.4 is why fusion past hydrogen pays

KEY RESULT

the two lines

$$B = \Delta m\, c^2$$

$$1\ \text{u} = 931.5\ \text{MeV}$$

B/A: 1.1 (d) → 7.1 (He) → 8.8 (Fe) → 7.6 (U).

INSIGHT

why the peak

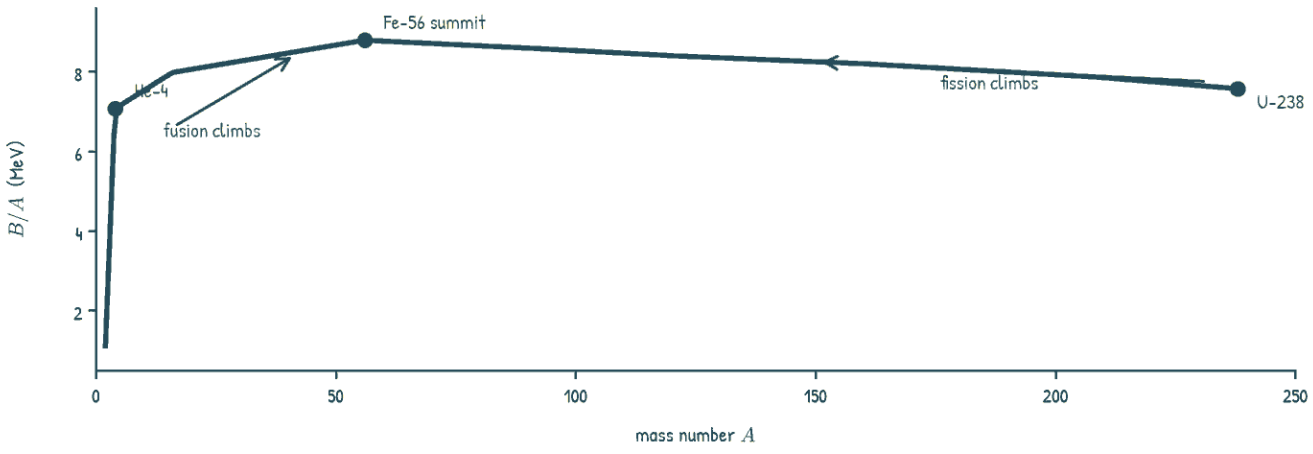
Short-range attraction grows as A, Coulomb repulsion as Z². Below iron the surface tax dominates; above it the Coulomb tax does.

TRAP

the sign

Using bare proton masses instead of hydrogen atoms in a beta problem, or calling the mass defect a loss of nucleons.

Any process that moves nucleons up the B/A curve releases the difference; iron is the summit



KEEP / B = Δm c², 1 u = 931.5 MeV, and B/A peaks at iron near 8.8 MeV

Why does the valley of stability bend?

? Light nuclei want equal protons and neutrons. Heavy ones want extra neutrons. The liquid drop says why.

1 Five terms, five physical reasons

Volume buys binding for every nucleon; surface taxes the ones at the edge; Coulomb punishes every proton pair; asymmetry punishes N ≠ Z because the two Fermi seas must be filled to the same height; pairing rewards even numbers.

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(N-Z)^2}{A} + \delta$$

→ typical coefficients 15.8, 18.3, 0.71, 23.2, 12 MeV

2 Maximise over Z at fixed A

Only the Coulomb and asymmetry terms care about Z once A is fixed. Setting the derivative to zero gives the valley floor: the most stable Z for that mass number.

$$Z_0 = \frac{A}{2 + \frac{a_c}{2a_a} A^{2/3}} = \frac{A}{2 + 0.0154 A^{2/3}}$$

→ A = 100 gives 42.9; A = 238 gives 92.0 — uranium sits on its own floor

3 Off the floor, beta decay is the traffic

A nucleus with too many neutrons converts one into a proton; too few, the reverse. The valley is the map and beta decay is the traffic that moves every nucleus toward the floor.

$$\beta^- : n \to p + e^- + \bar{\nu}_e \qquad \beta^+ : p \to n + e^+ + \nu_e$$

→ A never changes in beta decay; only Z does

4 A worked valley floor

Most stable charge for A = 100 and A = 238.

$$Z_0(100) = \frac{100}{2 + 0.0154(100)^{2/3}} = \frac{100}{2.33} = 42.9$$

$$Z_0(238) = \frac{238}{2 + 0.0154(238)^{2/3}} = 92.0$$

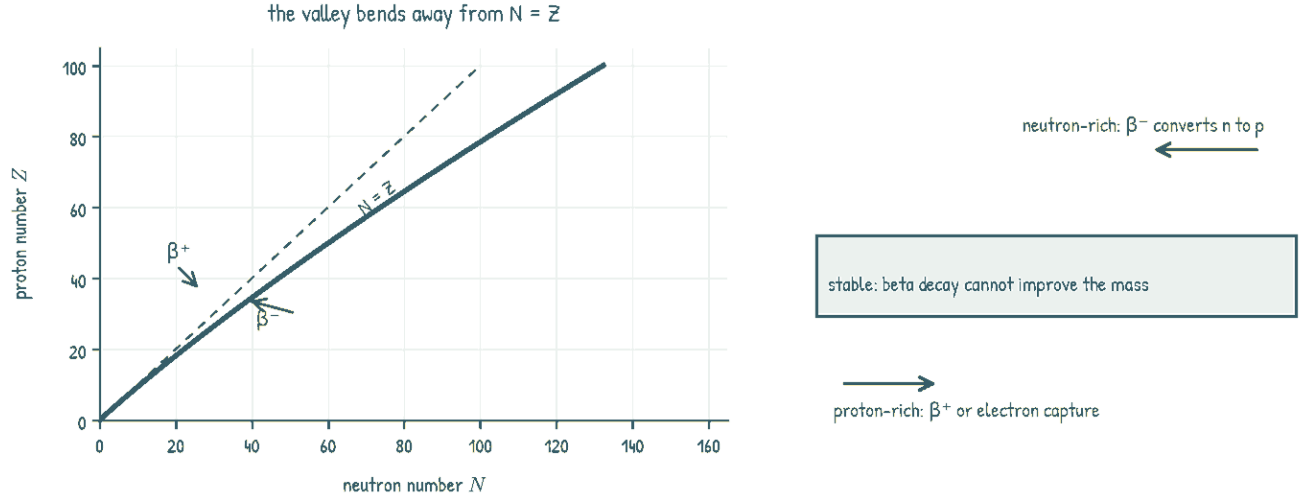
→ molybdenum-ruthenium for the first, uranium itself for the second

KEY RESULT
the floor
$$Z_0 = \frac{A}{2 + 0.0154 A^{2/3}}$$
Light nuclei: N ≈ Z. Heavy: N/Z ≈ 1.5.

GATE
liquid drop only
Shell closures dent and bump the floor — the model is the trend, not the detail.
Beyond bismuth no configuration is fully stable.

TRAP
which way
Neutron-rich nuclei go β^- (Z up). Proton-rich go β^+ or EC (Z down). Getting this backwards is the most common slip.

Coulomb grows as Z², surface as A^(2/3), so heavy nuclei need extra neutrons



KEEP / Coulomb grows as Z², surface as A^2/3, so Z0 = A/(2 + 0.0154 A^2/3)

Why does the alpha carry almost all the energy?

? Two bodies leave a decay at rest. Momentum must cancel, so the light one takes the lion's share.

1 Q is one subtraction

For any decay, Q = (mass in − mass out)c². Positive Q means it can happen; it says nothing about how fast. With atomic masses the electrons cancel, so the arithmetic is one line.

$$Q = (M_{\text{parent}} - M_{\text{daughter}} - M_{\alpha})c^2$$

→ alpha decay of U-238: Q = 4.27 MeV, which is why it can happen at all

2 Momentum splits Q inversely to mass

The parent was at rest, so the two products leave back to back with equal momentum. Kinetic energy goes as p²/2m, so the lighter body takes the larger share — the same rule as a recoiling gun.

$$K_{\alpha} = \frac{A-4}{A}Q, \qquad K_{\text{daughter}} = \frac{4}{A}Q$$

→ U-238: 4.20 MeV to the alpha, 0.07 MeV to the recoiling thorium

3 Discrete energies, because levels are discrete

The alpha can leave the daughter in its ground state or in an excited one; each branch has its own Q, so the alpha spectrum is a set of sharp lines, not a continuum. Any leftover excitation leaves later as a gamma.

$$Q_{\text{branch}} = Q_0 - E_{\text{ex}}, \qquad E_{\gamma} = E_{\text{ex}}$$

→ this is the nuclear analogue of an atomic line spectrum

4 A worked alpha split

Polonium-210 decays with Q = 5.40 MeV.

$$K_{\alpha} = \frac{206}{210}(5.40) = 5.30 \text{ MeV}, \qquad K_d = 0.10 \text{ MeV}$$

$$\sqrt{2m_{\alpha}K_{\alpha}} = \sqrt{2m_dK_d}$$

→ the daughter recoils with 2 % of the energy and a hundred times less speed

KEY RESULT
the split

$$Q = (m_{\text{in}} - m_{\text{out}})c^2$$

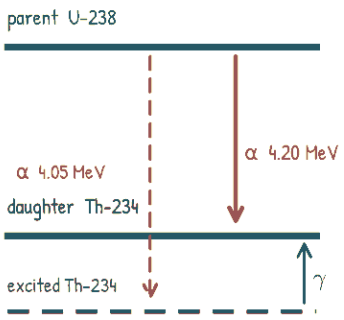
$$K_{\alpha} = \frac{A-4}{A}Q$$

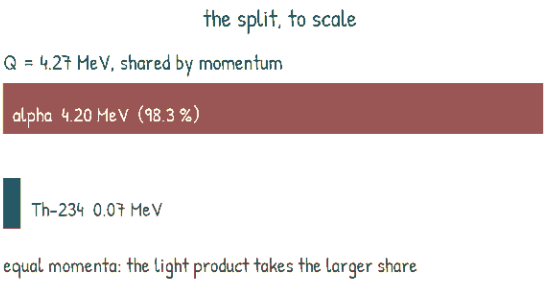
Lighter product takes the larger share.

GATE
two bodies only
The formula assumes the parent was at rest and only two products leave.
Beta decay is three bodies — hence its continuous spectrum.

TRAP
what Q is not
Q is not the alpha kinetic energy (multiply by (A−4)/A), and a positive Q does not mean an instant decay.

Q from masses; momentum splits it as (A−4)/A to the alpha and 4/A to the daughter





KEEP / Q from masses, then K_alpha = (A−4)Q/A — momentum fixes the split

0.1 eV 1 MeV 10 MeV

How can a beta spectrum be continuous?

? Alpha lines are sharp. Beta electrons arrive with every energy from zero to an endpoint. Something is missing.

1 Three bodies, not two

In beta decay the nucleus emits an electron and an antineutrino. With three bodies leaving, the energy can be shared in a continuum of ways: every electron energy from zero up to the endpoint Q is allowed, and the neutrino takes the remainder.

$$n \to p + e^- + \bar{\nu}_e \qquad p \to n + e^+ + \nu_e$$

→ the endpoint E_0 = Q is where the neutrino gets nothing

2 The electron rule, in one line each

Atomic masses include the electrons, and that changes the bookkeeping between the two beta signs. For β^- the daughter has one more electron, which the atomic mass already counts. For β^+ the positron is an extra particle and the atom has one electron too few.

$$Q_{\beta^-} = (M_P - M_D)c^2, \qquad Q_{\beta^+} = (M_P - M_D - 2m_e)c^2$$

→ C-14: 0.156 MeV. N-13: 2.220 − 1.022 = 1.20 MeV

3 Gamma and conversion finish the job

The daughter often stays excited. It can emit a photon of the level gap (gamma), or hand the same energy to an orbital electron (internal conversion), which then leaves with the gap minus its binding. Both carry the same information: the nuclear level diagram.

$$E_\gamma = E_{\text{ex}}, \qquad K_e = E_{\text{ex}} - B_{\text{shell}}$$

→ long-lived excited states are isomers; they decay later with the same energy

4 Two worked endpoints

Carbon-14 and nitrogen-13, with the electron rule applied.

$$Q_{\beta^-} = (14.003242 - 14.003074)(931.5) = 0.156 \text{ MeV}$$

$$Q_{\beta^+} = (13.005739 - 13.003355 - 0.001098)(931.5) = 1.20 \text{ MeV}$$

→ the 2m_e term is 1.022 MeV — larger than most beta Q-values, which is why β^+ needs a big parent

KEY RESULT
the two rules

$$Q_{\beta^-} = (M_P - M_D)c^2$$

$$Q_{\beta^+} = (M_P - M_D - 2m_e)c^2$$

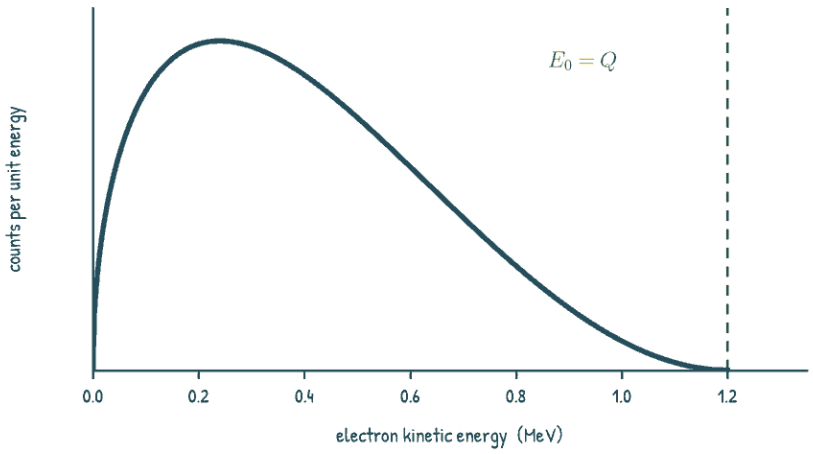
A is unchanged in both.

GATE
spectrum shape
The endpoint is exact; the shape near zero depends on the Coulomb pull on the emitted electron.
Neutrino mass is ~1 eV, invisible at MeV resolution.

TRAP
the classic trio
Giving the electron all of Q. Changing A by one. Forgetting the 2m_e for β^+.

A third particle carries the rest; the endpoint is where it carries nothing

N-13: every electron energy is allowed



two particles share Q



the gap between them is the missing energy

KEEP / a third particle carries the rest; the endpoint is where it carries nothing

0.1 eV 1 MeV 10 MeV

What does a half-life actually measure?

? Each nucleus has a fixed chance per second of dying. From that one assumption the whole calculus follows.

1 A memoryless probability gives an exponential

Assume every nucleus has the same chance λ dt of decaying in the next dt, however long it has already lived. Then dN = −λN dt integrates to an exponential, and the same λ governs the activity, because activity is λ times the number present.

$$N = N_0 e^{-\lambda t}, \qquad A_{ct} = \lambda N, \qquad T_{1/2} = \frac{\ln 2}{\lambda}$$

→ the log plot is a straight line of slope −λ; that is how half-lives are measured

2 Mean life is not half-life

The average lifetime is the integral of t over the decay distribution, and it comes out 1/λ — larger than the half-life by 1/ln2. The nucleus that survives longest is not the typical one.

$$\tau = \frac{1}{N_0}\int_0^{\infty} t\,(-dN) = \frac{1}{\lambda} = 1.443\,T_{1/2}$$

→ C-14: T½ 5730 y, τ 8270 y; Ra-226: 1600 y and 2310 y

3 Three things λ is not

λ does not depend on how many nuclei you have, on temperature, or on chemistry — with one exception: electron capture needs a bound electron, so extreme ionisation can shift it.

$$\lambda = \lambda_1 + \lambda_2 + \ \ldots\ , \qquad f_i = \frac{\lambda_i}{\lambda}$$

→ two decay modes add their constants; the branching fraction is the ratio

4 A worked working day

Tc-99m (T½ = 6 h) prepared at 800 MBq at 08:00; Co-60 mass for 1 Ci.

$$A(12\ \text{h}) = 800 \times 2^{-2} = 200\ \text{MBq}$$

$$N = \frac{3.7 \times 10^{10}}{4.17 \times 10^{-9}} = 8.9 \times 10^{18} \ \Rightarrow\ m = 0.88\ \text{mg}$$

→ a curie of a years-scale isotope is a speck; a curie of radium is a gram

KEY RESULT

the three

$$N = N_0 e^{-\lambda t}$$

$$A_{ct} = \lambda N$$

$$\tau = 1/\lambda = 1.443 T_{1/2}$$

INSIGHT

why λ is stable

Nuclear decay is a tunnelling or weak process at the MeV scale; room temperature is 25 meV. The ratio is 4×10⁷, so nothing ordinary can move it. Halving the sample halves A but leaves λ untouched.

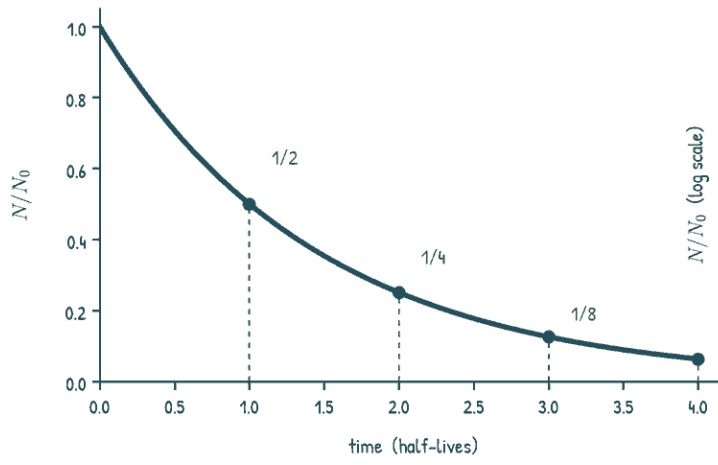
TRAP

the missing ln 2

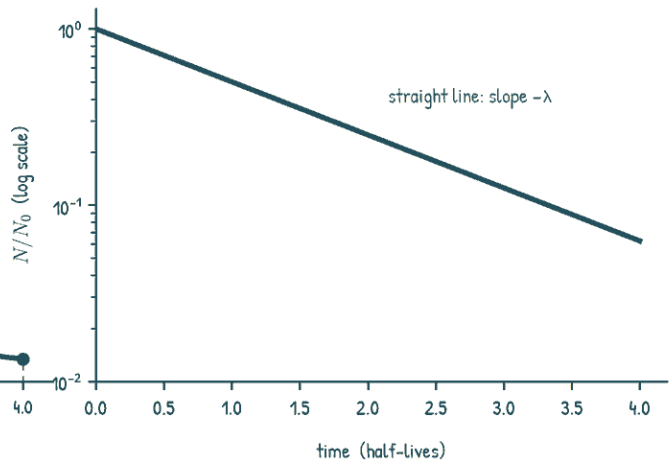
Writing e^{−t/T½} for e^{−λt}: at one half-life that gives 37 %, not 50 %. The exponent must carry ln 2.

N = N0 e^(−λt); the halving ladder for whole half-lives, the log line for the rest

each half-life halves the remainder



the log plot is the working tool



KEEP / N = N0e^{−λt}, T½ = ln2/λ, τ = 1.443 T½ — and λ is intensive

How do you date a bone?

? A living thing keeps its radiocarbon topped up. When it dies the clock starts, and the ratio is the hand.

1

The clock is an exchange that stops

Cosmic rays keep the atmospheric $^{14}C/^{12}C$ ratio roughly fixed. Living tissue exchanges carbon, so it matches the atmosphere; death stops the exchange and the ^{14}C already inside decays away.

$$\frac{A}{A_0} = e^{-\lambda t} \;\Rightarrow\; t = \frac{T_{1/2}}{\ln 2}\ln\frac{A_0}{A}$$

→ 10 disintegrations per minute per gram against 15 gives 3350 years

2

The range is set by counting, not by physics

After ten half-lives one nucleus in a thousand is left, and counting statistics dominate. That is why the ceiling is about 50 000 years, and why rocks need longer-lived clocks.

$$t_{10} = 10(5730) = 5.7\times10^{4}\ \text{y}, \qquad \frac{\sigma}{N} = \frac{1}{\sqrt{N}}$$

→ K-Ar: 1.25 Gyr; U-Pb: 4.47 Gyr — same logarithm, different constant

3

The assumptions you must state

Constant production, a closed sample, and a known initial ratio. The last is the weak link, so tree rings calibrate the raw ratio; the isochron method removes it by plotting one ratio against another.

$$\frac{^{87}\text{Sr}}{^{86}\text{Sr}} = \left(\frac{^{87}\text{Sr}}{^{86}\text{Sr}}\right)_0 + \frac{^{87}\text{Rb}}{^{86}\text{Sr}}\left(e^{\lambda t}-1\right)$$

→ a straight line whose slope is the age — the intercept absorbs the unknown start

4

A worked carbon age

A bone gives 10 disintegrations per minute per gram; living tissue gives 15.

$$t = \frac{5730}{\ln 2}\ln\frac{15}{10} = 8267(0.405) = 3350\ \text{y}$$

$$\frac{A}{A_0} = 2^{-t/T_{1/2}} = 0.667$$

→ between one and two half-lives, which is the range where the method is most reliable

KEY RESULT

the clock

$$t = \frac{T_{1/2}}{\ln 2}\ln\frac{A_0}{A}$$

C-14: 5730 y, range to ~50 ka.

K-Ar, U-Pb for rock.

GATE

when it fails

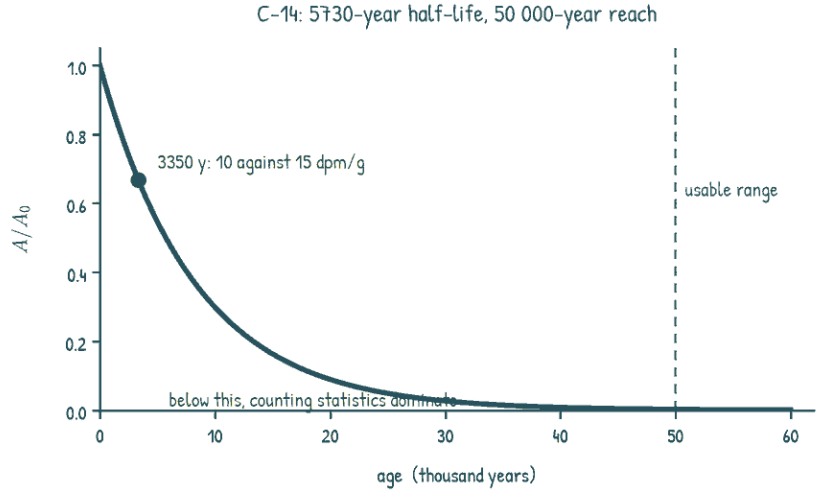
Open samples, contamination, fossil fuel (dead carbon), and dates near the limit where the count is a handful.

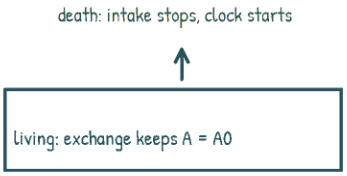
TRAP

the ratio inverted

Using A/A_0 instead of A_0/A flips the sign of the age. Check that a smaller activity gives a positive age.

The clock runs from the moment exchange stops; the range ends where the counts do





calibration curves correct the raw ratio

KEEP / $t = (T_{½}/\ln 2)\ln(A_0/A)$: three assumptions, one logarithm

0.1 eV 1 MeV 10 MeV

Why does a decay chain settle into a fixed ratio?

? A daughter fed by its parent has two clocks. When one is much slower, the other just follows.

1 Two clocks in one equation

The daughter is produced at rate $\lambda_1 N_1$ and destroyed at rate $\lambda_2 N_2$. Solving the pair with a daughter that starts at zero gives a rise-and-fall: the daughter activity climbs, peaks, then locks onto the parent's decay.

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2 \;\Rightarrow\; A_2 = \frac{\lambda_2}{\lambda_2 - \lambda_1} A_1 \left(1 - e^{-(\lambda_2 - \lambda_1)t}\right)$$

→ the bracket is the approach to equilibrium; after a few daughter half-lives it is one

2 Name the regime before doing arithmetic

If the parent is far longer-lived, the ratio settles at one: secular equilibrium, the state of old radium samples. If it is only somewhat longer-lived, the ratio settles at a number above one: transient equilibrium, and the chain decays with the parent's clock.

$$\text{secular}: \ \frac{A_2}{A_1} \to 1, \qquad \text{transient}: \ \frac{A_2}{A_1} \to \frac{\lambda_2}{\lambda_2 - \lambda_1}$$

→ $\lambda_2 = 3\lambda_1$ gives a ratio of 3/2; the daughter is always the more active one

3 Peak time and branching

The daughter activity peaks where its derivative vanishes; after that it inherits the parent's slope. If a nuclide has two decay modes, the constants add and each branch takes its share of the decays.

$$t_{\max} = \frac{\ln(\lambda_2/\lambda_1)}{\lambda_2 - \lambda_1}, \qquad \lambda = \lambda_a + \lambda_b, \quad f_a = \frac{\lambda_a}{\lambda}$$

→ the long-term tempo always belongs to the longest-lived member present

4 A worked generator

Parent $T_{1/2}$ = 18 h, daughter $T_{1/2}$ = 6 h: find the ratio and the peak.

$$\lambda_1 = 0.0385\ \text{h}^{-1}, \ \lambda_2 = 0.1155\ \text{h}^{-1}$$

$$\frac{A_2}{A_1} = \frac{0.1155}{0.0770} = 1.50, \qquad t_{\max} = \frac{\ln 3}{0.0770} = 14.3\ \text{h}$$

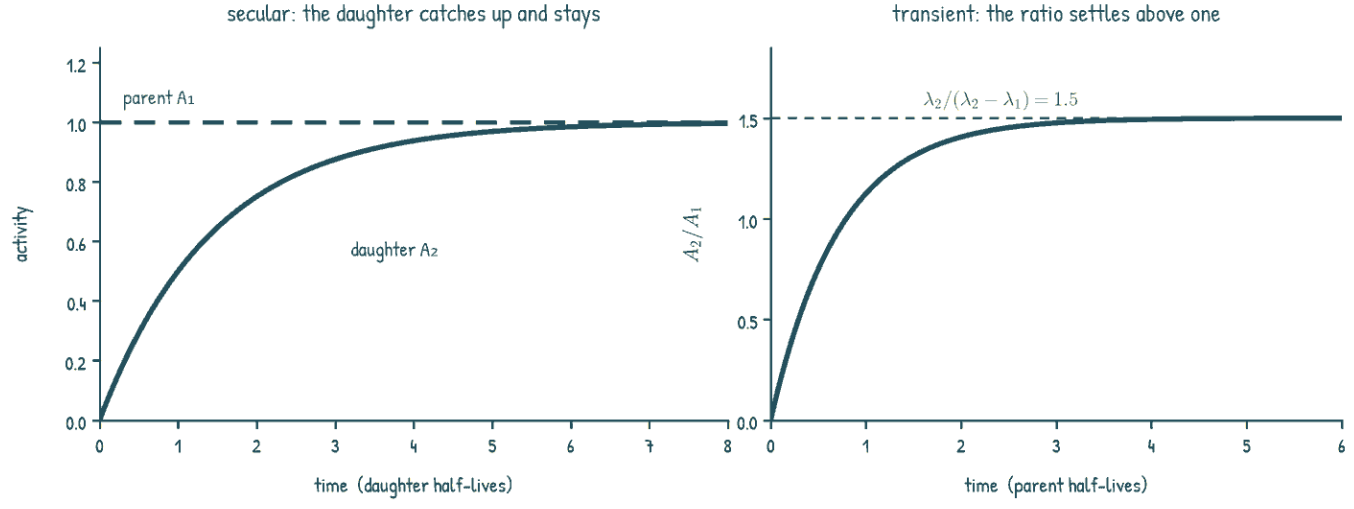
→ elute the daughter at 14 h, not at the end of the day

KEY RESULT
the regimes
Secular: parent ≫ daughter, ratio 1.
Transient: ratio $\lambda_2/(\lambda_2 - \lambda_1)$.
No equilibrium if the parent is the shorter one.

GATE
what the formulas assume
Closed system, no daughter at t = 0, and a single decay path for each nuclide.

TRAP
which is parent
Comparing half-lives backwards. The longer-lived isotope is the one that sets the chain's pace.

Secular equilibrium gives $A_2 = A_1$; transient gives $A_2/A_1 = \lambda_2/(\lambda_2 - \lambda_1)$



KEEP / secular: $A_2 = A_1$; transient: $A_2/A_1 = \lambda_2/(\lambda_2 - \lambda_1)$

Where does a gigawatt come from?

? One fission releases 200 MeV. Multiply by the fission rate and a gram of uranium outruns a tonne of coal.

1 The energy budget of one split

A thermal neutron leaves U-236 excited above its fission barrier; the drop necks and Coulomb wins. The fragments fly apart with most of the energy; the rest arrives later as neutrons, betas and gammas.

$$^{235}\mathrm{U} + n \rightarrow {}^{236}\mathrm{U}^{*} \rightarrow X + Y + 2n + 200\ \mathrm{MeV}$$

→ ≈165 MeV fragments, 5 MeV neutrons, 20 MeV betas and gammas (delayed), 10 MeV neutrinos lost

2 Convert to engineering units

Divide 200 MeV into a gram of U-235: 8.2×10¹⁰ J, about 3.4 tonnes of coal. A gigawatt thermal plant therefore burns about a kilogram a day.

$$\frac{10^{9}}{200 \times 1.602 \times 10^{-13}} = 3.1 \times 10^{19}\ \mathrm{s^{-1}}, \qquad \frac{8.6 \times 10^{13}}{8.2 \times 10^{10}} = 1.05\ \mathrm{kg/day}$$

→ the reactor's daily meal fits in a pocket; the waste does not

3 The chain needs a multiplication factor

k compares neutrons in one generation with the last. Below one the fire dies, at one it is steady, above one it rises. A reactor is designed to sit at k = 1 with delayed neutrons giving the control loop time to act; a bomb is prompt-critical by construction.

$$k < 1 \text{ subcritical}, \qquad k = 1 \text{ steady}, \qquad k > 1 \text{ rising}$$

→ moderator slows, control rods absorb, coolant carries heat, shielding guards the outside

4 A worked power balance

A 1 GW electric plant at 33 % efficiency.

$$P_{\mathrm{th}} = 3.0\ \mathrm{GW}: \quad \frac{3 \times 10^{9}}{200(1.602 \times 10^{-13})} = 9.4 \times 10^{19}\ \mathrm{s^{-1}}$$

$$m = \frac{3(1.05)}{1} = 3.2\ \mathrm{kg/day\ of\ U{-}235}$$

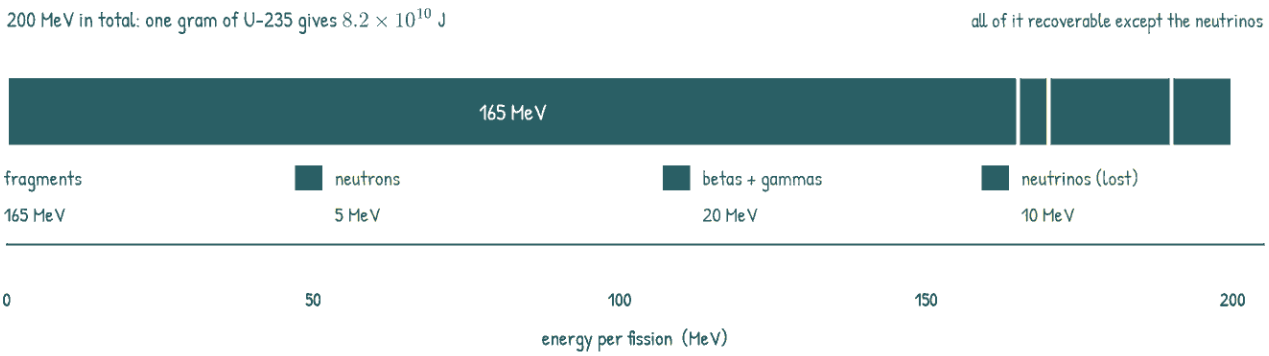
→ three tonnes of coal would do the same job — the factor 10⁴ in one line

KEY RESULT
the arithmetic
200 MeV per fission.
8.2×10¹⁰ J per gram of U-235.
3.1×10¹¹ fissions per second per GW thermal.

GATE
k and the reactor
A power reactor cannot detonate like a weapon: dilute fuel, delayed neutrons, negative feedbacks.
That is not a claim that accidents are impossible.

TRAP
the two units
Confusing thermal and electrical power (factor ≈3), and quoting 200 MeV per *nucleus* as if all the fuel fissioned.

Fragments carry most of it; the neutrinos leave with 10 MeV and are never recovered



KEEP / 200 MeV per fission: 8.2×10¹⁰ J per gram, 3.1×10¹¹ fissions per second per GW

0.1 eV — 1 MeV ▼ — 10 MeV

Why is the Sun not ten billion kelvin?

? Protons repel hard. Classical physics demands an impossible temperature; the Sun runs on the tail of a distribution.

1 The barrier is real and it is large

Two protons at 2 fm sit on a Coulomb hill of 0.72 MeV. Setting k_BT equal to it gives 8×10⁹ K — the classical answer, five hundred times the core.

$$U = \frac{ke^2}{2\ \text{fm}} = 0.72\ \text{MeV} \ \Rightarrow\ T = \frac{U}{k_B} \approx 8 \times 10^9\ \text{K}$$

→ that is the roll-over temperature, not the operating temperature

2 Tunnelling and the Maxwell tail meet in the middle

The reaction rate is the product of how many particles have energy E (a falling exponential) and how likely they are to tunnel through the barrier (a rising one). The product peaks at the Gamow energy, far above k_BT and far below the barrier.

$$E_0 = \left(\frac{E_G (k_B T)^2}{4}\right)^{1/3} \approx 6\ \text{keV} \quad \text{at } 15\ \text{MK}$$

→ the Sun burns at 1.5×10⁷ K because its particles tunnel, not because they climb

3 The chain, and what it pays

The p-p chain fuses four protons in three steps; the first is weak and slow, and it sets the Sun's clock. The 26.7 MeV appears as light and neutrinos.

$$4p \rightarrow {}^4\text{He} + 2e^+ + 2\nu + 26.7\ \text{MeV}, \qquad \dot{m} = \frac{L}{c^2} = 4.3 \times 10^9\ \text{kg/s}$$

→ only 0.7 % of the fused mass becomes energy — and that is still 4 million tonnes a second

4 A worked barrier

Deuterium plus tritium, and the classical temperature of the proton barrier.

$$Q = (0.018883\ \text{u})(931.5) = 17.6\ \text{MeV}$$

$$T = \frac{0.72\ \text{MeV}}{8.617 \times 10^{-11}\ \text{MeV/K}} = 8.4 \times 10^9\ \text{K}$$

→ the D-T reaction is the easiest one, and it still needs 100 MK in the lab

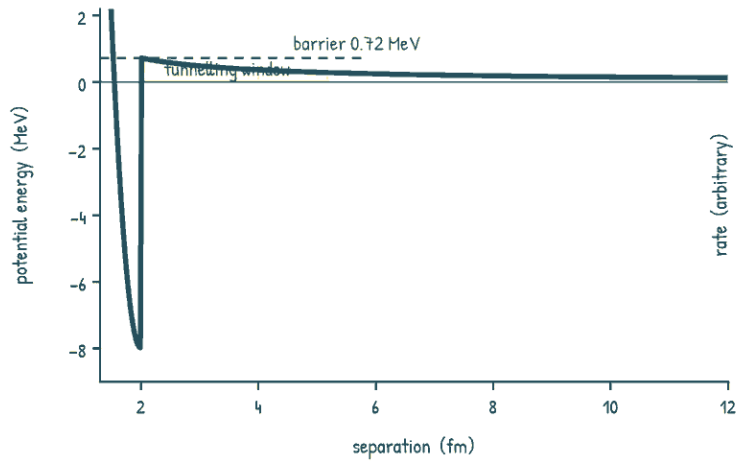
KEY RESULT
the three numbers
Barrier 0.72 MeV at 2 fm.
Core 1.5×10⁷ K, Gamow peak ~6 keV.
26.7 MeV per helium; 4.3×10⁹ kg/s lost.

GATE
when the chain applies
p-p dominates cool stars; the CNO cycle takes over above ~1.7×10⁷ K with carbon as catalyst.
Both give the same net energy.

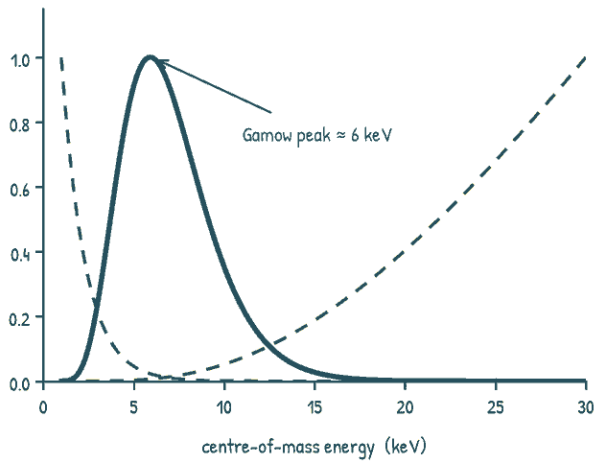
TRAP
the one-step Sun
Four protons do not collide at once. Treating the chain as a single reaction loses the weak first step that sets the lifetime.

Classically the Sun needs 8×10⁹ K; tunnelling and the Maxwell tail make 1.5×10⁷ K enough

the Coulomb hill, and the well behind it



tail × tunnel = the working window



KEEP / the Coulomb barrier needs 8×10⁹ K classically; tunnelling plus the Maxwell tail makes 1.5×10⁷ K enough